

한 가지.

Def.

1) $y = h(x)$

$$h(x) = \frac{f(x)}{g(x)}$$

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

2) $y = \frac{1}{g(x)}$

or $y = \frac{1}{g(x)}$

$$y' = -\frac{g'(x)}{g(x)^2}$$

P.f. 1)

$$\left[\frac{f(x)}{g(x)} \right]' = \left[f(x) \cdot \frac{1}{g(x)} \right]'$$

곱의 미분.

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$= \left(f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left[\frac{1}{g(x)} \right]' \right)$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \left(-\frac{g'(x)}{g(x)^2} \right)$$

$$= \frac{f'(x)}{g(x)} - f(x) \cdot \frac{g'(x)}{g(x)^2}$$

$$= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

$$\therefore \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

P.f. 2)

$$\left[\frac{1}{g(x)} \right]' = \lim_{\Delta x \rightarrow 0} \frac{\frac{1}{g(x+\Delta x)} - \frac{1}{g(x)}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{g(x) - g(x+\Delta x)}{g(x+\Delta x)g(x)}}{\Delta x} = - \lim_{\Delta x \rightarrow 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} \cdot \frac{1}{g(x+\Delta x)g(x)}$$

$$g'(x) =$$

$$\lim_{\Delta x \rightarrow 0} \frac{g(x+\Delta x) - g(x)}{\Delta x}$$

$$\lim_{\Delta x \rightarrow 0} \frac{1}{g(x+\Delta x)g(x)}$$

한 가지
성질

$$\lim_{\Delta x \rightarrow 0} [f(x)g(x)] = f(x) \cdot g(x)$$

$$\therefore \frac{d}{dx} \left(\frac{1}{g(x)} \right) = -\frac{g'(x)}{g(x)^2}$$

$$= -\frac{g'(x)}{g(x)^2}$$

$$= \lim_{\Delta x \rightarrow 0} \Delta x \rightarrow 0 : \frac{1}{g(x)g(x)} = \frac{1}{g(x)^2}$$